

# Detailed Lecture Notes: Maximum Likelihood Estimation (MLE)

**Video:** [Maximum Likelihood, clearly explained!!!](#)

**Channel:** StatQuest with Josh Starmer

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## 1. The Goal of Maximum Likelihood Estimation (MLE)

- **Problem:** We have a set of observed data (e.g., mouse weights), and we need to estimate the **parameters** of the entire population that produced that data.
  - **Goal of MLE:** To find the optimal **parameters** (like the Mean and Standard Deviation) for a chosen **statistical distribution** that make the observed data **most likely** to have occurred.
  - **Benefit of Fitting a Distribution:** It makes the data easier to work with, more general, and applicable to broader types of experiments.
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## 2. Choosing and Parameterizing the Distribution

- **Choice:** We first choose a distribution type (e.g., **Normal Distribution** is often chosen for continuous data like weights, provided it looks symmetrical and centered).
  - **Parameters:** To fully define a Normal Distribution, we need two parameters:
    - i. **Mean ( $\mu$ ):** Determines the **location** or center of the distribution.
    - ii. **Standard Deviation ( $\sigma$ ):** Determines the **shape** (how spread out) the distribution is.
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## 3. The Process of Maximizing Likelihood

MLE is a process of testing different parameter values to find the ones that maximize the likelihood of the observed data.

### A. Maximizing the Likelihood for the Mean ( $\mu$ )

1. **Testing a Value:** Start with a test value for the distribution's mean ( $\mu$ ).
2. **Calculating Likelihood:** The **Likelihood** is the combined **probability density (height of the curve)** at all the observed data points for that specific test distribution.
  - If the distribution's mean is far from the observed data, the likelihood is **low**.
3. **Shifting and Comparing:** We systematically **shift the distribution's mean ( $\mu$ )** and calculate the likelihood at each new position.
4. **Finding the Maximum:** The value of  $\mu$  that results in the **highest likelihood** is the Maximum Likelihood Estimate (MLE) for the mean.
5. **Result for Normal Distribution:** The MLE for the distribution's Mean is equal to the **Mean of the observed data**.

### B. Maximizing the Likelihood for the Standard Deviation ( $\sigma$ )

- Once the optimal Mean ( $\mu$ ) is fixed, the process is repeated for the Standard Deviation ( $\sigma$ ).
  - We try different  $\sigma$  values (different curve widths/shapes) and choose the one that, when combined with the optimal  $\mu$ , **maximizes the likelihood** of observing the data.
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## 4. Terminology Distinction

- **Likelihood** in statistics specifically refers to the situation where you have **fixed data** and you are finding the **optimal distribution parameters** to fit that data.
- The result of MLE is finding the single distribution (defined by the estimated parameters) that best explains your collected sample data.

